Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere
An Autonomous Institute Affiliated to VTU
Department of CSE/ISE and Allied Branches

Course Title	Data Structures and Applications	**Course Code**	BCSPCC302
Course Instructor	Dr. Bhuvaneshwari K V, Dr. Gururaj T, Prof. Puneeth SP, Prof. Kotramma T S, Prof. Anusha N	**Semester-Section**	III
IA Test No.	I	**Max. Marks**	30
Date	14.11.2025	**Time**	3.00 to 04.15PM

Course Outcome Statements: After the successful completion of the course, the students will be able to

CO1	Apply arrays and structures for efficient data representation and memory management.
CO2	Design solutions for data organization and problem-solving using stack and queues.
CO3	Implement various linked list operations for efficient data storage and manipulation.
CO4	Develop solutions using trees and graphs to model real-world problems.
CO5	Utilize advanced data structures like hashing and priority queues for optimized data retrieval and management.

Note: Answer any one full question from each Part.

Q. No.	Question	Marks	RBT Level	CO
	Part A			
1	a) Define Data Structures. **Describe** classification of Data Structures and **explain** their common operations.	6	L2	1
	b) Define Polynomial. **Develop** a C function for addition of two polynomial. Illustrate with an example.	6	L3	1
	OR			
2	a) List and **explain** dynamic memory functions with syntax and example.	6	L2	1
	b) Define pattern matching. **Construct** failure table for the pattern P=ababd. **Apply** KMP algorithm to match the pattern P in the given input text T=ababcabcabababd	6	L3	1
	Part B			
3	a) Define Stack. **Outline** the different operations that can be performed on stack with suitable C functions.	6	L2	2
	b) i) **Convert** the given infix expression to postfix expression using stack A+(B*C-(D/E^F)*G)+H ii) **Evaluate** the given Postfix expression using stack 546+*493/+*	6	L3	2
	OR			
4	a) Define Queue. **Develop** a C function to implement insertion, deletion and display operations.	6	L3	2
	b) **Build** a C function for dynamic implementation of push() and stackfull() operation using dynamic array.	6	L3	2
	Part C			
5	Define Linked list. **Explain** different type of linked list with neat diagram.	6	L2	3
	OR			
6	**Develop** a C function for the following operations on singly linked list: i) Insertion at the beginning ii) Deletion at the end iii) Display all the elements.	6	L3	3

RBT (Revised Bloom's Taxonomy) Levels : Cognitive Domain					
L1 : Remembering	L2 : Understanding	L3 : Applying	L4 : Analysing	L5 : Evaluating	L6 : Creating

Course Coordinator **DAAC Coordinator** **Program Coordinator (ISE)** **BOE chairman**

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Course Title	Data Structures and Applications	**Course Code**	BCSPCC302
Course Instructor	Dr. Sreenivasa B R, Dr. Naveen Kumar K R, Dr. Varsha M, Prof. Vishwanath, Prof. Gowramma H, Prof. Sushma C, Prof. Rachana G Sunkad	**Semester-Section**	III
IA Test No.	II	**Max. Marks**	30
Date	08.12.2025	**Time**	3:00 to 4:15 PM

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Apply arrays and structures for efficient data representation and memory management.
CO2	Design solutions for data organization and problem-solving using stacks and queues.
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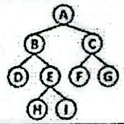
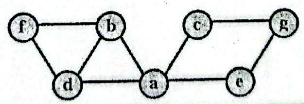
<table>
<tr><th>Q. No.</th><th></th><th>Question</th><th>Marks</th><th>RBT Level</th><th>CO</th></tr>
<tr><td></td><td></td><td>Part A</td><td></td><td></td><td></td></tr>
<tr><td>1</td><td>A</td><td>Write recursive C functions for in-order, pre-order, and post-order traversals of a binary tree. Also, find all the traversals for the given tree.

</td><td>6</td><td>L3</td><td>4</td></tr>
<tr><td></td><td>B</td><td>Define a Binary Search tree. Construct a binary search tree (BST) for the following elements: 100, 85, 45, 55, 120, 20, 70, 90, 115, 65, 130, 145. Traverse using in-order, pre-order, and post-order traversal techniques.</td><td>6</td><td>L3</td><td>4</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>2</td><td>A</td><td>Write a C function to traverse a graph using DFS. Also, show the BFS traversal for the following graph.

</td><td>6</td><td>L3</td><td>4</td></tr>
<tr><td></td><td>B</td><td>Define Selection tree. Construct a min winner tree for the runs of a game given below. Each run consists of the players' values. Find the first five winners.
<table>
<tr><td>10</td><td>9</td><td>20</td><td>6</td><td>8</td><td>9</td><td>90</td><td>17</td></tr>
<tr><td>15</td><td>20</td><td>20</td><td>15</td><td>15</td><td>11</td><td>95</td><td>18</td></tr>
<tr><td>16</td><td>38</td><td>30</td><td>25</td><td>50</td><td>16</td><td>99</td><td>20</td></tr>
<tr><td></td><td></td><td></td><td>28</td><td></td><td></td><td></td><td></td></tr>
</table></td><td>6</td><td>L3</td><td>4</td></tr>
<tr><td></td><td></td><td>Part B</td><td></td><td></td><td></td></tr>
<tr><td>3</td><td>A</td><td>Define hashing. Discuss different hashing functions with an example.</td><td>6</td><td>L2</td><td>5</td></tr>
<tr><td></td><td>B</td><td>Define collision. List the methods used to resolve collisions. Construct a hash table ht of size 15 (using open addressing method) to store the following words: like, a, tree, you, first, find, a, place, to, grow, and, then, branch, out.</td><td>6</td><td>L3</td><td>5</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>4</td><td>A</td><td>Write a short note on i) Priority queues ii) Optimal binary search tree.</td><td>6</td><td>L2</td><td>5</td></tr>
</table>

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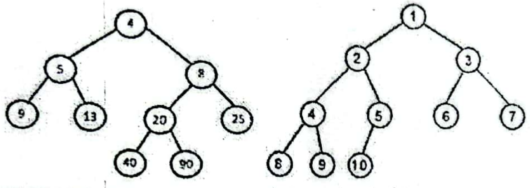
Note: Answer any one full question from each Part.

Q. No.		Question	Marks	RBT Level	CO
4	B	**Define a leftist tree. Check whether the given binary tree is a leftist tree or not. Justify your answer.** 	6	L3	5
		Part C			
5	A	**Write the function for the following operation on a Doubly Linked List (DLL).** 1. Inserting a node at the front end. 2. Deleting a node from the rear end.	6	L3	3
		OR			
6	B	**Define Sparse matrix. For the given sparse matrix, give the linked list representation:** $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 5 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 8 \end{bmatrix}$	6	L3	3

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Course Coordinator | 31/12/25 **DAAC Coordinator** | **Program Coordinator** | **BoE** 31/12/2025